

		Resistor across R and S is $(0.00 + 10.0 + 10.0) \Omega$. Hence the balanced length should be 3 times that of 5.0Ω , or 37.5 cm .
24	A	Magnetic force = BIL $1.0 = (0.50)(I)(2.0)$ $I = 1.0 \text{ A}$ Using Fleming's Left Hand Rule, Since resulting force acts into the plane of the paper (thumb points into the plane) and magnetic field (first finger points to the right), current flows from R to S (upwards).
25	B	Initially, when the beam of electrons passes through without deviation, this means that the electric force (upward) is equal to the magnetic force